

PRABAL SINGH

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PROFESSIONAL SUMMARY

Junior network and cloud infrastructure engineer at SJSU (CNSM, GPA 3.6) with hands-on experience in wireless compliance testing and Python/Bash automation from a UL Solutions internship; incoming NDE Intern at Amazon (Summer 2026). Proficient in TCP/IP, VLANs, DNS/DHCP, OSPF, BGP, firewalls, and IPv4/IPv6; experienced with Linux, Cisco CLI, AWS (EC2, S3, IAM), and Docker; holds CCST with completed CCNA coursework; targeting cloud and AI infrastructure roles in solutions architecture.

EDUCATION

Bachelor of Science, Engineering Technology

Concentration: Computer Network System Management (CNSM)

May 2027

San Jose State University, San Jose, CA, GPA: 3.6

Minor: Business, Computer Science

SKILLS

Networking & Security: TCP/IP, UDP, DNS, DHCP, VLANs, Subnetting, NAT, IPv4/IPv6, OSPF, VPNs, Firewalls, BGP

Programming: Python, Java, SQL, Bash, Pandas, Web Scraping, Data Analysis, Configuration scripting

Cloud & Virtualization: AWS EC2, S3, IAM, Cloud Networking Basics, Virtual Machines, Containers (Docker)

Systems & Tools: Linux, Windows, macOS, Cisco CLI, Git/GitHub, Wireshark, Raspberry Pi, Arduino, REST APIs

WORK EXPERIENCE

Network Development Engineer Intern, Amazon – Arlington, VA

May 2026 – Aug 2026

- Building an AI-powered diagnostic agent using Amazon Bedrock to automate TOR switch build failure analysis and resolution for network infrastructure.

Peer Academic Success Coach, UHS SJSU

Aug 2024 - Present

- Provide academic advising and holistic support to 100+ first-year students, helping them navigate college life and build academic goals.
- Design and implement engagement programs, boosting residential community involvement by 30% through targeted workshops and events.
- Earned 95% positive feedback in student satisfaction surveys, recognized for consistent availability, clear communication, and personalized academic support.

Testing Engineer Intern, UL Solutions

May 2025 - Aug 2025

- Conducted RF compliance testing (Wi-Fi, BLE, LTE) for 30+ product modules, validating power levels and communication stability against FCC/ISED standards to support client product certifications.
- Automated test data collection workflows using Python and Bash, reducing manual test time by ~40% and improving result reproducibility across wireless RF evaluations.
- Supported SAR test accuracy by performing dielectric liquid verification and system calibration checks, directly contributing to compliance report quality and client certification timelines.

CERTIFICATIONS

- Cisco Certified Support Technician (CCST) – Networking

Dec 2025

- CCNA: Enterprise Networking, Security, and Automation - Cisco Networking Academy

Aug 2025

PROJECT EXPERIENCE

Hear the Light – LED Sound Level Meter, SJSU

May 2025

- Designed and built a sound-level meter using Arduino and LEDs to visualize ambient noise levels.
- Integrated microphone sensor with Arduino Uno to capture sound input and programmed real-time LED responses.
- Enhanced understanding of embedded systems, sensors, and signal processing.

Line Following Robot, SJSU

Dec 2023

- Developed an autonomous line-following robot with Arduino, using infrared sensors for precision tracking.
- Integrated hardware and coding to create a line-following robot achieving over 95% accuracy in sensor-based tracking.
- Demonstrated expertise by programming and calibrating sensors to operate on varied surfaces and speeds, enhancing functionality and control by 40%.

ACTIVITIES

Network Engineering Technology Society (NETS), Member

Jan 2025 - Present

- Participated in workshops and events focused on networking, cloud systems, and cybersecurity; collaborated with peers on technical projects.